Layer-Local Implicit Token Lenses Recover Multi-Token Words

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Abstract

Standard logit lenses decode intermediate language-model states into explicit tokenizer items, but recent evidence suggests that models also build word-like representations that need not correspond to a single token. We ask whether a lens can expose these implicit vocabulary items directly. We instantiate implicit tokens as 128 lowercased English word types from WIKITEXT-2 that QWEN2.5-0.5B tokenizes into multiple pieces. For each layer, we build a layer-local implicit vocabulary table by averaging normalized hidden states from training occurrences, then decode held-out occurrences by cosine similarity to the word prototypes. On 512 held-out examples, the validation-selected ITL at layer 5 reaches 19.53% top-1 accuracy, 25.59% top-5 accuracy, and 0.240 MRR. This exceeds a final-subtoken majority baseline by 7.03 percentage points (bootstrap 95% CI [2.53, 11.72], paired randomization $p = 0.0045$) and chance accuracy by 18.75 points. It also beats the strongest raw final-prototype implicit baseline by 5.47 points (bootstrap 95% CI [2.15, 8.79]). A Tuned-Lens-style ridge translator to final-layer prototypes performs poorly at this data scale, reaching only 3.71% top-1. These results show that a supervised, prototype-based implicit-token lens can recover whole-word identity beyond subtoken leakage, while leaving open unsupervised discovery and causal validation of implicit vocabulary items.

1 Introduction

Logit lenses are useful because they translate hidden states into something readers can inspect: a distribution over vocabulary items. That output space is also their main constraint. If a model has composed several tokenizer pieces into a word-like internal representation, a standard lens can still only name explicit tokenizer IDs.

This constraint matters for modern language models because tokenization often splits words, names, and expressions into pieces whose meanings do not match the full item. Recent work on token erasure argues that representations at the final token of a multi-token expression rapidly lose component-token information, leaving a footprint of a higher-level implicit item [Feucht et al., 2024]. Related inner-lexicon work finds that LLMs reconstruct whole-word representations from subword sequences and can use such representations for vocabulary expansion [Kaplan et al., 2025]. These results raise a direct question for lens methods: can we decode into the implicit item itself, rather than into its last tokenizer piece?

Existing tools do not fully answer this question. TUNED LENS learns affine translators that make intermediate states decodable, but its output space remains the model's explicit vocabulary [Belrose et al., 2023]. Patchscopes can elicit natural-language descriptions of hidden states, but they do not define a fixed vocabulary table that can be reused for retrieval [Ghandeharioun et al., 2024]. Prompt and prefix tuning show that continuous vectors can act like virtual tokens [Lester et al., 2021, Li and Liang, 2021, Liu et al., 2022], but they are usually evaluated through task performance rather than through direct decoding of latent lexical items.

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Multi-token words grouped by final subtoken	→	Layer states final subtoken position	→	Implicit vocabulary cosine retrieval over words

Figure 1: Overview of the ITL experiment. We define implicit tokens as whole word types that are absent as single tokenizer entries, build layer-local prototype rows from training occurrences, and retrieve held-out word identity from intermediate hidden states.

We propose an Implicit Token Lens (ITL): a logit-lens-style decoder whose output rows are prototypes for multi-token words. Figure 1 shows the setup. We collect occurrences of multi-token word types, store hidden states at the final subtoken position, and average training states into one normalized prototype row per word and layer. At evaluation time, the lens scores a held-out hidden state against all implicit-token rows by cosine similarity.

Our main result is positive but bounded. On Qwen2.5-0.5B and local WikiText-2, a validation-selected layer-local ITL recovers held-out identity among 128 multi-token words with 19.53% test top-1 accuracy. This is above final-subtoken majority at 12.50%, majority-word chance at 0.78%, and the strongest raw final-prototype baseline at 14.06%. The paired improvement over final-subtoken majority is 7.03 percentage points with bootstrap 95% CI [2.53, 11.72]. The same ITL also beats the raw final-prototype baseline by 5.47 points with bootstrap 95% CI [2.15, 8.79]. At the same time, a ridge translator to final-layer prototypes fails, reaching only 3.71% top-1 accuracy, so the evidence favors layer-local prototype tables rather than a universal translated output space in this small-data setting.

Our contributions are:

- We propose a prototype implicit-token output space for logit-lens-style decoding of multi-token word representations.
- We construct a leakage-controlled vocabulary of 128 multi-token words with 16 final-subtoken groups and 8 words per group.
- We compare ITL against final-subtoken, compositional embedding, raw final-prototype, and ridge-to-final baselines with paired statistical tests.
- We analyze layer-wise behavior and representative errors, showing that word-level decoding peaks early while explicit final-subtoken identity is strongest at the embedding layer.

The rest of the paper is organized as follows. Section 2 positions the experiment relative to vocabulary-space lenses and implicit lexicon work. Section 3 defines the ITL, dataset, baselines, and metrics. Section 4 reports quantitative results and error analysis. Section 5 discusses what this experiment does and does not establish, and section 6 concludes.

2 Related Work

Vocabulary-space lenses. The standard logit lens projects intermediate activations through the final unembedding, making hidden states readable as next-token distributions. Tuned Lens improves this idea by learning a layer-specific affine translator before decoding into the vocabulary [Belrose et al., 2023]. Future Lens extends hidden-state decoding toward later tokens rather than only the immediate next token [Pal et al., 2023]. These methods show that hidden states contain decodable information, but their categories are still explicit tokenizer IDs or future explicit tokens. Our experiment keeps the lens-style retrieval interface but changes the output space to whole multi-token word labels.

Implicit lexical representations. Token erasure provides direct motivation for an implicit-token output space. Feucht et al. [2024] find that last-token representations for multi-token words and named entities lose component-token information early, and they use this erasure pattern as evidence for implicit vocabulary items. Kaplan et al. [2025] similarly argue that LLMs perform intrinsic detokenization, reconstructing whole-word representations from subword inputs in early and middle layers. Work on transformer feed-forward layers as key-value memories also supports the idea that model components can retrieve concept-like information through directions that are not simply surface token IDs [Geva et al., 2021]. We build on this line by asking whether a fixed table of whole-word prototypes can decode held-out occurrences quantitatively.

Table 1: Experimental setup. The vocabulary is balanced so that each final-subtoken group contains eight word types.

Component	**Value**
Model	QWEN2.5-0.5B ('Qwen/Qwen2.5-0.5B')
Dataset	Local raw WIKITEXT-2
Candidate implicit vocabulary	128 lowercased alphabetic multi-token words
Final-subtoken control	16 groups, 8 words per final tokenizer piece
Splits	10 train, 2 validation, 4 test occurrences per word
Total examples	2,048 (1,280 train; 256 validation; 512 test)
Hidden states	25 levels including embeddings; hidden size 896
Extraction	Final subtoken position, max length 96, batch size 64
Hardware	One NVIDIA RTX A6000

Table 2: Example implicit vocabulary groups. All words in a row share the same final tokenizer piece, so final-subtoken identity alone cannot choose among the eight candidates.

Final token	**Candidate words**
'land'	england, ireland, maryland, queensland, portland, lowland, finland, highland
'on'	xenon, galveston, simon, eaton, algernon, baron, luzon, lebanon

Natural-language inspection and virtual tokens. Patchscopes use activation patching and target prompts to make hidden states inspectable through generated language [Ghandeharioun et al., 2024]. This is more expressive than a vocabulary projection, but it is less directly a reusable table of categories. Continuous prompt methods provide a complementary precedent: prompt tuning, prefix tuning, and P-Tuning learn vectors that function like tokens outside the ordinary tokenizer matrix [Lester et al., 2021, Li and Liang, 2021, Liu et al., 2022]. AutoPrompt searches for discrete prompt tokens instead [Shin et al., 2020]. These approaches show that token-like behavior is not limited to static vocabulary rows. Our focus is narrower: we do not learn task prompts, but decode naturally occurring hidden states into a supervised implicit vocabulary.

Model and corpus. We run the experiment on QWEN2.5-0.5B, part of the Qwen2.5 family [Qwen Team, 2025], using local raw WIKITEXT-2 data derived from the WikiText language modeling corpus [Merity et al., 2016]. The setup is intentionally small enough to cache all hidden states and to evaluate paired differences on the same held-out occurrences.

3 Methodology

Problem setup. Let $\mathcal{V}_I$ be a fixed implicit vocabulary of multi-token word types. For an occurrence i of word $y_i \in \mathcal{V}_I$, we feed the context ending at that word into a frozen causal language model and extract the hidden state $\boldsymbol{h}_{\ell,i} \in \mathbb{R}^d$ at the final subtoken position for layer ℓ. The task is whole-word retrieval: rank every candidate $w \in \mathcal{V}_I$ and recover y_i on held-out occurrences. This is different from ordinary next-token decoding because the correct label is a current whole word, not the next explicit tokenizer ID.

Candidate vocabulary and data. We use local raw WIKITEXT-2 and QWEN2.5-0.5B. Candidate words are lowercased alphabetic word types that tokenize into at least two pieces under the Qwen tokenizer. To reduce final-subtoken leakage, we arrange the 128 candidates into 16 final-subtoken groups with 8 words per group. Therefore, knowing only the final tokenizer ID gives at most one correct word out of eight within a group under the balanced evaluation split. Each word contributes 10 training, 2 validation, and 4 test occurrences, for 2,048 total examples. Table 1 summarizes the experimental setup, and table 2 shows two candidate groups.

Layer-local implicit token lens. The primary ITL builds one implicit vocabulary table per layer. For every word w and layer ℓ, we average normalized training states and normalize the resulting

Table 3: Validation-selected whole-word retrieval results. Higher is better for all metrics. Best comparable values in each metric column are in **bold**.

Method	Selected layer	Validation top-1	Test top-1	Test top-5 / MRR
IMPLICIT TOKEN LENS	5	**21.09%**	**19.53%**	25.59% / **0.240**
FINAL-SUBTOKEN MAJORITY	–	12.50%	12.50%	– / –
RAW-FINAL	22	11.33%	14.06%	**29.69%** / 0.220
MEAN-SUBTOKEN	0	8.59%	10.94%	21.88% / 0.175
RIDGE-TO-FINAL ITL	17	6.25%	3.71%	11.52% / 0.092
MAJORITY WORD	–	0.78%	0.78%	– / –

row:

$$\boldsymbol{p}_{\ell,w} = \frac{\frac{1}{|\mathcal{D}_w|}\sum_{i\in\mathcal{D}_w}\frac{\boldsymbol{h}_{\ell,i}}{\|\boldsymbol{h}_{\ell,i}\|_2}}{\left\|\frac{1}{|\mathcal{D}_w|}\sum_{i\in\mathcal{D}_w}\frac{\boldsymbol{h}_{\ell,i}}{\|\boldsymbol{h}_{\ell,i}\|_2}\right\|_2}. \tag{1}$$

At validation or test time, a held-out state is scored by cosine similarity,

$$s_\ell(i,w) = \frac{\boldsymbol{h}_{\ell,i}}{\|\boldsymbol{h}_{\ell,i}\|_2}^\top \boldsymbol{p}_{\ell,w}. \tag{2}$$

The predicted implicit token is $\arg\max_{w\in\mathcal{V}_I} s_\ell(i,w)$. We select the ITL layer by validation top-1 accuracy and report the corresponding test metrics.

Baselines and ablations. We compare against five controls. MAJORITY WORD always predicts the most common training word and gives the 1/128 chance rate under balanced labels. FINAL-SUBTOKEN MAJORITY uses only the target occurrence's final tokenizer ID and predicts the most frequent training word in that suffix group. MEAN-SUBTOKEN averages each candidate word's input embedding rows and scores hidden states against those compositional vectors. RAW-FINAL builds prototypes from final-layer training states only, then compares every layer directly to this fixed final-layer table. RIDGE-TO-FINAL ITL trains a ridge-regularized affine map, with an unregularized intercept, from each layer's training states into the final-layer prototype space; translated states are normalized before scoring against final prototypes. We also report an explicit final-subtoken embedding diagnostic over 16 final-token IDs. This diagnostic measures recoverability of the suffix token, not whole-word decoding, so we do not treat it as a competing whole-word method.

Metrics and statistical tests. We report top-1 accuracy, top-5 accuracy, and mean reciprocal rank (MRR) for whole-word retrieval. All primary comparisons use validation-selected layers and the same 512 test occurrences. We estimate accuracy and paired accuracy-difference intervals with 3,000 bootstrap samples. We also run paired randomization tests with 6,000 samples and McNemar exact tests for top-1 correctness disagreements.

Implementation details. The model is 'Qwen/Qwen2.5-0.5B'. Hidden states include the embedding level and 24 transformer layers, for 25 hidden-state levels with hidden size 896. During extraction, we apply the model's final normalization module to each layer state, normalize vectors, and store final-position hidden states. The run used batch size 64, maximum sequence length 96, seed 42, and one NVIDIA RTX A6000. Key package versions were Python 3.12.8, PyTorch 2.12.1+cu130, Transformers 5.12.1, Datasets 5.0.0, NumPy 2.5.0, SciPy 1.18.0, and scikit-learn 1.9.0.

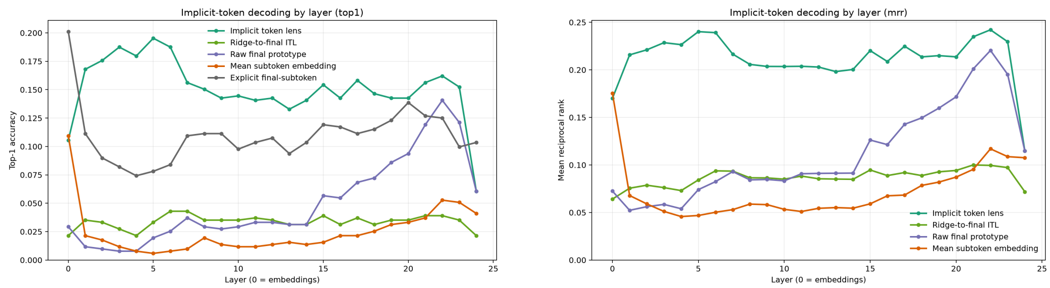
4 Results

Main comparison. Table 3 reports validation-selected layers and held-out test performance. The layer-local ITL selects layer 5 and reaches 19.53% test top-1 accuracy, compared with 12.50% for FINAL-SUBTOKEN MAJORITY and 0.78% for MAJORITY WORD. Among learned whole-word methods, ITL also has the best validation top-1, test top-1, and test MRR. RAW-FINAL reaches a higher test top-5 accuracy at its selected layer, but its top-1 accuracy remains 5.47 points below ITL.

Statistical evidence. Table 4 shows paired test-set comparisons. The primary comparison is ITL layer 5 versus FINAL-SUBTOKEN MAJORITY. The accuracy difference is +7.03 percentage points with bootstrap 95% CI [2.53, 11.72], paired randomization $p = 0.0045$, and McNemar exact $p =$

Table 4: Paired test-set comparisons for top-1 accuracy. Differences are ITL minus the comparison method. Confidence intervals are paired bootstrap intervals.

Comparison	**Accuracy diff.**	**Bootstrap 95% CI**	**Randomization p**	**McNemar p**
ITL layer 5 vs. FINAL-SUBTOKEN MAJORITY	+7.03 pp	[2.53, 11.72] pp	0.0045	0.0046
ITL layer 5 vs. RAW-FINAL layer 22	+5.47 pp	[2.15, 8.79] pp	0.0023	0.0020
ITL layer 5 vs. MEAN-SUBTOKEN layer 0	+8.59 pp	[5.08, 12.11] pp	< 0.001	< 0.001



(a) Top-1 accuracy

(b) Mean reciprocal rank

Figure 2: Layer-wise decoding results on validation and test splits. ITL peaks early at layer 5, while the raw final-prototype baseline is strongest late in the model. The final-subtoken diagnostic measures suffix-token recovery rather than whole-word retrieval.

0.0046. The selected ITL also beats RAW-FINAL layer 22 by +5.47 points and MEAN-SUBTOKEN layer 0 by +8.59 points, with positive confidence intervals in both cases.

Layer-wise behavior. Figure 2 shows how retrieval quality changes by layer. ITL rises quickly and peaks at layer 5, then declines gradually. RAW-FINAL peaks much later, at layer 22, which suggests that final-layer prototypes are geometrically best matched to late states. The explicit final-subtoken diagnostic is strongest at layer 0, with 20.12% test accuracy over the 16 suffix IDs, and falls in early and middle layers while word-level ITL improves. This pattern is qualitatively consistent with token erasure: surface suffix identity becomes less directly recoverable as a whole-word representation becomes more decodable.

Ablation: translating to final prototypes is weak. The ridge-to-final ablation is the closest analogue to the TUNED LENS idea in this experiment: map each layer into a fixed final-layer prototype space, then decode against that table. It performs poorly, selecting layer 17 by validation top-1 and reaching only 3.71% test top-1. This does not show that affine translators cannot work in principle. It does show that with 10 training occurrences per word and 896-dimensional states, the simple ridge translator is a poor fit compared with layer-local prototype rows.

Error analysis. Table 5 gives representative failures. Many mistakes stay inside the same final-subtoken group, such as ‘england’ decoded as ‘highland’ or ‘queensland’. However, not all errors are suffix-only: one ‘ireland’ occurrence is ranked second as ‘lebanon’. The failures suggest that the hidden state mixes whole-word identity with residual orthographic information and contextual semantics. This is also why the final-subtoken control is important: ITL improves over suffix identity, but suffix and topic remain substantial confounds.

5 Discussion

What the result supports. The experiment supports a narrow, testable claim: a supervised layer-local table of implicit-token prototypes can recover held-out multi-token word identity above strong controls. The improvement over FINAL-SUBTOKEN MAJORITY is important because each suffix group contains eight candidate words with the same final tokenizer piece. The effect is therefore not explained by simply reading out the last subtoken. The improvement over MEAN-SUBTOKEN also suggests that the signal is not only a compositional average of input embeddings.

Why layer-local prototypes matter. The strongest method is not the ridge-to-final translator. Instead, each layer needs its own prototype table. This points to a practical lesson for implicit-token

Table 5: Representative ITL layer-5 errors on held-out examples. Some failures preserve the final tokenizer piece, while others reflect contextual or semantic confusions beyond the suffix group.

Target	Prediction	Rank	Shared final token
england	highland	10	yes ('land')
ireland	lebanon	2	no
england	queensland	114	yes ('land')

lenses: the implicit item may be linearly accessible in a layer's local geometry even when a small-data affine map into a final-layer space is unreliable. This is compatible with prior work showing that word-level information emerges in early and middle layers [Kaplan et al., 2025], while the final representation is optimized for next-token prediction rather than current-word naming.

Limitations. The implicit vocabulary is supervised and fixed to known word types. The experiment does not discover arbitrary latent vocabulary entries. It uses one small open model, one English corpus, and occurrence-level splits from WIKITEXT-2, so article and domain correlations may help prototypes. Candidate words are frequent enough for 16 occurrences and are balanced into final-subtoken groups; rarer words, nonwords, named entities, and phrases may behave differently. The method also establishes decodability, not causality. We do not show that the model relies on the decoded prototype for downstream prediction. Finally, the ridge failure may reflect sample size, dimensionality, and regularization rather than a fundamental weakness of translated implicit-token spaces.

Broader implications. Tokenizer-bound interpretability tools can miss categories that the model internally treats as larger units. A reusable implicit vocabulary table is one way to test for those categories without asking the model to verbalize every hidden state. If scaled and made less supervised, this could help compare tokenizers, diagnose detokenization failures, initialize vocabulary expansion, and localize where word, entity, or phrase representations become stable.

6 Conclusion

We introduced an IMPLICIT TOKEN LENS that decodes hidden states into whole multi-token word types rather than explicit tokenizer IDs. On QWEN2.5-0.5B and WIKITEXT-2, a layer-local prototype table recovers held-out word identity at 19.53% top-1 accuracy over 128 candidates, beating final-subtoken majority, mean subtoken embeddings, and raw final-layer prototypes with paired statistical support.

The key takeaway is that implicit-token decoding is feasible in a supervised prototype setting, but the geometry is layer-specific. The next step is to remove some supervision: mine candidate implicit items from token-erasure footprints, validate them with Patchscopes, and test causal interventions that replace or ablate an implicit word prototype. Scaling the same evaluation to larger Qwen and Llama-family models, named entities, phrases, and non-English data will show how general the implicit-token lens really is.

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